Structural Classification of Undecidable Problems via Non-Middle Fallacy and Dynamic Normalization

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Abstract

Many central open problems in modern mathematics exhibit a common structural difficulty: they attempt to determine static truth values for phenomena that are inherently dynamic, inter-structural, or self-referential. In this paper, we introduce the notion of a *Non-Middle Fallacy* (NMF), a structural obstruction arising when a static formal system attempts to enforce absolute equality across incompatible logical regimes.

We provide a unified classification of major undecidable or controversial problems—including the ABC conjecture, the Riemann hypothesis, P versus NP, the Continuum Hypothesis, and Godel-type incompleteness— according to distinct types of NMF. We then propose a dynamic normalization framework, denoted $\overset{\text{mw}}{=}$, which replaces static equality with a stability-based equivalence. This framework is based on the Sunyata-Rupa Duality Theory (SKDT) and does not claim to produce classical proofs, but rather to normalize structural tension and classify the logical status of statements beyond the theorem/hypothesis dichotomy.

Finally, we introduce a refined taxonomy of mathematical truth (Nomological Truth, Practical Truth, and Quasi-Theorem), intended to clarify the epistemic status of results that are structurally stable yet classically undecidable.

1 Introduction

1.1 Motivation

A remarkable feature of modern mathematics is that many of its most important problems resist resolution not because of technical difficulty alone, but because of deeper structural constraints. Examples include the ABC conjecture, the Riemann hypothesis, the P versus NP problem, and the Continuum Hypothesis. Despite decades of effort, these problems remain undecided within standard formal frameworks.

At the same time, foundational results such as Gödel's incompleteness theorems and Cohen's independence results demonstrate that undecidability is not an anomaly but an intrinsic feature of sufficiently expressive formal systems. This raises a natural question:

> Are these difficulties isolated phenomena, or manifestations of a common structural pattern?

This paper argues for the latter.

1.2 Overview of the Approach

We introduce the concept of a *Non-Middle Fallacy* (NMF), a structural configuration in which a static formal system attempts to enforce absolute equality or decidability across domains that require dynamic mediation. Rather than treating undecidability as a failure, we treat it as a diagnostic signal of structural mismatch.

Our approach proceeds in three stages:

1. Define NMFs as structural obstructions arising from static formalization.
2. Classify major undecidable or controversial problems according to distinct types of NMF.
3. Introduce a dynamic normalization framework, $\overset{\mathrm{mw}}{=}$, based on the SKDT (Sunyata-Rupa Duality Theory), that replaces static equality with stability-based equivalence.

The goal is not to refute classical mathematics, but to extend its descriptive power by clarifying the structural status of problems that lie beyond classical proof.

2 Non-Middle Fallacy (NMF)

2.1 Definition

We define a *Non-Middle Fallacy* (NMF) as follows.

Definition 1 (Non-Middle Fallacy). An NMF occurs when a static formal system attempts to enforce a binary, absolute decision (true/false, equal/unequal) on a phenomenon whose logical structure is inherently dynamic, inter-structural, or self-referential, thereby producing undecidability, inconsistency, or persistent controversy.

NMFs do not indicate an error in reasoning, but rather a mismatch between the expressive mode of the formal system and the structure of the problem.

2.2 Structural Origins of NMF

We identify three recurring structural sources of NMFs.

1. **Dynamic vs Static Mismatch:** Finite static descriptions are applied to infinite or process-oriented phenomena.
2. **Inter-Structural Comparison:** Distinct logical domains are compared via static equality.
3. **Boundary and Self-Reference:** A system attempts to fully describe or decide properties of its own expressive boundary.

These sources give rise to distinct classes of NMF, which we analyze in the next sections.

3 Classification of Non-Middle Fallacy

In this section, we present a unified classification of Non-Middle Fallacy based on their structural origin. Although individual problems differ greatly in content, their logical pathologies exhibit strong structural similarities.

3.1 Inter-Structural Comparison NMF

3.1.1 Description

An *Inter-Structural Comparison NMF* arises when two logically distinct structures are compared using static equality. The comparison itself is well-defined syntactically, but lacks a stable semantic interpretation.

Typical examples involve comparisons between:

- additive and multiplicative structures,
- discrete and continuous domains,
- computational processes and verification procedures.

3.1.2 Representative Problems

Problem	Status	Structural Description
ABC conjecture	Undecided / disputed	Static comparison between additive relation $a + b = c$ and multiplicative radical $rad(abc)$.
Riemann hypothesis	Undecided	Static comparison between discrete prime distribution and analytic continuation of zeta zeros.
Goldbach conjecture	Undecided	Additive decomposition constrained by multiplicative (prime) structure.
P vs NP	Undecided	Equality between dynamic computation time and static verification time.
IUT treatment of ABC	Controversial	Comparison across meta-arithmetic universes with restricted identification rules.

3.1.3 Structural Analysis

In all such cases, the difficulty does not lie in the absence of local arguments, but in the lack of a coherent static notion of equality that bridges the two domains. Attempts to resolve these problems classically tend to increase structural complexity rather than reduce logical tension.

3.2 Boundary and Self-Reference NMF

3.2.1 Description

A *Boundary/Self-Reference NMF* occurs when a formal system attempts to decide properties that inherently refer to its own expressive limits. This type of NMF is characterized by unavoidable meta-level escalation.

3.2.2 Representative Problems

Problem	Status	Structural Description
Godel incompleteness	Proven undecidable	Finite axiomatic system attempts to decide its own consistency.
Chaitin incompleteness	Proven undecidable	Finite description attempts to capture irreducible algorithmic information.
Russell paradox	Resolved by stratification	Unrestricted self-membership within a static definition.

3.2.3 Structural Analysis

These results are often interpreted negatively as limitations. From the NMF perspective, they instead indicate that static closure is the wrong objective. The failure is not logical inconsistency, but the absence of a dynamic stabilization mechanism.

3.3 Hierarchical Inclusion NMF

3.3.1 Description

A *Hierarchical Inclusion NMF* arises when an infinite generative hierarchy is forced into a static axiomatic determination. This is typical in set-theoretic questions concerning infinity.

3.3.2 Representative Problems

Problem	Status	Structural Description
Continuum Hypothesis	Independent of ZFC	Static decision of infinite cardinal hierarchy.
Large cardinal axioms	Independent of ZFC	Existence claims beyond fixed axiomatic reach.

3.3.3 Structural Analysis

Independence results show that these questions do not admit a privileged static resolution. Any answer requires extending the universe of discourse, confirming their classification as NMF phenomena rather than incomplete proofs.

3.4 Unified Perspective

Despite surface differences, all NMF types share a common pattern: a static binary decision is imposed on a structure that demands dynamic mediation. This observation motivates the search for a normalization framework that replaces static equality with a stability-based criterion.

4 Dynamic Normalization via $\overset{\text{mw}}{=}$ (SKDT Framework)

This section introduces the core technical contribution of this paper: the replacement of static equality by a dynamic stability operator, denoted by $\overset{\text{mw}}{=}$, within the framework of the Sunyata-Rupa Duality Theory (SKDT).

4.1 Limitations of Static Equality

Classical equality is defined as a timeless, context-independent relation. While effective in homogeneous domains, it becomes unstable when applied across structurally distinct systems, leading to NMF.

Formally, expressions of the form

$$A = B$$

fail when the underlying generative rules or semantic resolution are incompatible.

4.2 Definition of the $\overset{\text{mw}}{=}$ Operator based on SKDT

We define $\overset{\text{mw}}{=}$ as a relation between structures that captures dynamic convergence rather than static identity, utilizing the SKDT framework.

Definition 4.1 (Middle-Way Equivalence ($\overset{\text{mw}}{=}$))**.** Let S be the Rupa (form/phenomenon) Category and K be the Sunyata (void/potential) Category. Let $F : \mathsf{S} \to \mathsf{K}$ and $G : \mathsf{K} \to \mathsf{S}$ be the inter-structural transformations (functors of 'Pratityasamutpada' or Interdependence).

We write a structure $X \in \mathsf{S}$ is Middle-Way Equivalent to a structure $Y \in \mathsf{S}$,

$$X \overset{\text{mw}}{=} Y$$

if the distortion measure E associated with the transformation loop satisfies:

$$E(G \circ F(X)) \to E_{min}$$

where E_{min} is the minimal admissible structural tension, often $E_{min} = 0$ but not necessarily. Stability is defined by convergence to this minimal admissible value, not by exact coincidence.

4.3 Finite Stabilization Principle

A central principle of $\overset{\text{mw}}{=}$-based mathematics is the exclusion of static infinity, which is the primary source of Hierarchical Inclusion NMF.

Principle 1 (Finite Stabilization)**.** All valid constructions must admit a finite stabilization process. Infinite objects are replaced by arbitrarily extensible but finite procedures whose stability can be evaluated dynamically.

This principle eliminates the need for the axiom of infinity and replaces it with iterative convergence criteria.

4.4 Resolution of Inter-Structural Comparison NMF

The $\overset{\text{mw}}{=}$ framework replaces static comparison with a convergence condition.

4.4.1 Functorial Coherence

Let $S_1, S_2 \in \mathsf{S}$ be two distinct structures (e.g., additive and multiplicative arithmetic). The condition

$$f : S_1 \to S_2, \quad g : S_2 \to S_1$$

The dynamic coherence is maintained if the composition stabilizes:

$$g \circ f \overset{\text{mw}}{=} id_{S_1}$$

This replaces static comparison with a convergence condition that can be verified algorithmically.

4.5 Resolution of Boundary and Self-Reference NMF

Boundary-related NMF are resolved by abandoning static completeness and introducing dynamic reflection.

4.5.1 Meta-Level Stabilization

Let C be a formal system (an object in S) and M be its reflective meta-system (a higher object in S, perhaps S's self-reference). Instead of requiring C to decide its own consistency, we introduce a reflective transformation

$$R : C \to M$$

such that instability in C triggers corrective normalization in M.

Consistency is thus maintained dynamically as a fixed point of iteration, rather than a statically provable property.

4.6 Resolution of Hierarchical Inclusion NMF

Hierarchical questions (e.g., CH) are reformulated under $\overset{\text{mw}}{=}$ as:

> Does a given generative process stabilize under finite extension?

Since the axiom of infinity is excluded, the notion of fixed infinite cardinality is replaced by the notion of admissible generative processes. If the generative process does not stabilize, the question of its static size is declared structurally invalid rather than undecidable.

4.7 Summary

The $\overset{\text{mw}}{=}$ operator provides a low-complexity alternative to classical meta-theoretic escalation. By replacing equality with dynamic stability, it resolves all major classes of Non-Middle Fallacy while preserving computational and conceptual tractability.

5 New Classes of Mathematical Truth

This section introduces a reclassification of mathematical truth based on dynamic coherence rather than static provability. The classical dichotomy of theorem versus conjecture is shown to be insufficient in the presence of Non-Middle Fallacy.

5.1 Limitations of the Theorem–Conjecture Dichotomy

In classical mathematics, a statement is assigned one of two statuses: **Theorem** or **Conjecture**. This classification presupposes that all meaningful statements admit static truth values. However, NMF demonstrates that many foundational questions fail not because of missing techniques, but because the question itself is structurally ill-posed.

5.2 Dynamic Truth and Coherence Levels

Under the $\overset{\text{mw}}{=}$ framework, truth is evaluated by the degree to which a statement maintains dynamic stability across structural transformations.

We therefore introduce three classes of truth, ordered by increasing strength of coherence.

5.3 Nomological Truth

Definition 5.1 (Nomological Truth)**.** A statement is called a *nomological truth* if it expresses a structural constraint required for universal dynamic stability. Such a statement resolves all associated NMF and remains coherent under arbitrary admissible transformations ($\overset{\mathrm{mw}}{=}$-coherence level: Universal).

Nomological truths are not contingent on a particular axiom system. They correspond to invariants of the middle-way normalization process.

Example. The reformulated ABC principle, expressed as a bound on structural distortion between additive and multiplicative constructions, is a candidate for nomological truth.

5.4 Practical Truth

Definition 5.2 (Practical Truth)**.** A statement is called a *practical truth* if it guarantees dynamic stability under finite operational procedures, even if universal coherence is not established ($\overset{\mathrm{mw}}{=}$-coherence level: Operational).

Practical truths are characterized by effective convergence rather than absolute invariance.

Example. Monte Carlo methods exemplify practical truth: finite sampling does not establish exact equality, but reliably converges toward stable outcomes.

5.5 Quasi-Theorem

Definition 5.3 (Quasi-Theorem)**.** A statement is called a *quasi-theorem* if it is internally consistent and provable within a newly constructed formal universe, but lacks verified coherence with external or classical systems ($\overset{\mathrm{mw}}{=}$-coherence level: Local).

This class legitimizes innovative frameworks without prematurely forcing classical equivalence.

Example. Results derived within inter-universal Teichmueller theory are best understood as quasi-theorems until functorial coherence with classical arithmetic is demonstrated.

5.6 Structural Classification of Major Problems

The following table summarizes the proposed classification.

Problem	NMF Type	Truth Class
ABC conjecture	Inter-structural	Nomological
Riemann hypothesis	Inter-structural	Nomological
P vs NP	Dynamic/static	Nomological
Monte Carlo methods	Dynamic/static	Practical
IUT results	Inter-structural	Quasi-theorem
Continuum hypothesis	Hierarchical	Invalid question

5.7 Consequences for Mathematical Practice

This reclassification shifts the goal of mathematics from absolute closure to controlled coherence. Innovation is no longer blocked by premature demands for static equivalence, while foundational confusion is reduced by explicit recognition of NMF.

6 Conclusion

This paper has argued that many of the most persistent problems in mathematics arise not from technical difficulty, but from a shared structural defect, the Non-Middle Fallacy (NMF). Static formal systems attempt to force dynamic, relational, or self-referential phenomena into fixed logical identities, thereby producing undecidability, paradox, or endless controversy.

6.1 Summary of Results

The main contributions of this work are as follows.

1. A unified definition of NMF was introduced, explaining undecidability, incompleteness, and foundational conflict as a single structural phenomenon.
2. The operator $\stackrel{\text{mw}}{=}$, grounded in the SKDT framework, was proposed as a dynamic replacement for static equality, allowing comparisons based on convergence and stability rather than absolute identity.
3. A normalization framework based on finite operations and dynamic middle-way coherence was shown to dissolve major classes of NMF.
4. Classical open problems were reclassified according to their structural nature rather than their proof status.
5. Three new truth classes were introduced: nomological truth, practical truth, and quasi-theorem.

These results suggest that the limits identified by Gödel and later formal results do not merely constrain provability, but signal the need for a change in the underlying notion of mathematical truth.

6.2 Implications for Foundations

Under the proposed framework, mathematics is no longer required to be self-contained in a static sense. Instead, it is understood as a dynamically stabilized system whose coherence is maintained through iterative normalization.

This perspective resolves several long-standing tensions:

- Incompleteness becomes a structural boundary condition rather than a failure.
- Independence results are reinterpreted as indicators of invalid static questions.
- Highly complex theories are evaluated by coherence rather than by forced equivalence with classical formalisms.

In particular, inter-universal constructions may be meaningful without serving as classical proofs, provided their internal dynamics are stable.

6.3 Outlook

The framework developed here opens multiple directions for future research.

- In number theory, major conjectures may be reformulated as nomological truths expressing unavoidable structural constraints.
- In logic and computer science, dynamic normalization provides a new lens on complexity, verification, and computation.
- In physics and information theory, middle-way coherence offers a foundation for understanding stability without singularities.

More broadly, this work suggests that the role of mathematics is shifting from the pursuit of absolute identities to the cultivation of stable relations. The adoption of dynamic middle-way principles may therefore mark a transition comparable in significance to the axiomatization movement of the twentieth century.

6.4 Final Remark

The Non-Middle Fallacy is not an obstacle to be eliminated, but a signal that a static formulation has exceeded its domain of validity. By embracing dynamic coherence through $\overset{\mathrm{mw}}{=}$, mathematics can proceed beyond its classical limits without abandoning rigor.

This shift does not weaken mathematics. It clarifies its scope, strengthens its foundations, and restores contact between formal reasoning and the structures it seeks to understand.